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Deductibles in health insurance: can the actuarially fair premium reduction exceed the deductible?

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Abstract

The actuarially fair premium reduction in case of a deductible relative to full insurance is affected by: (1) out-of-pocket payments, (2) moral hazard, (3) administrative costs, and, in case of a voluntary deductible, (4) adverse selection. Both the partial effects and the total effect of these factors are analyzed. Moral hazard and adverse selection appear to have a substantial effect on the expected health care costs above a deductible but a small effect on the expected out-of-pocket expenditure. A premium model indicates that for a broad range of deductible amounts the actuarially fair premium reduction exceeds the deductible. © 2000 Elsevier Science Ireland Ltd. All rights reserved.

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1. Introduction

Both theoretical and empirical studies suggest that health insurance with some cost sharing is ‘optimal’ for the insured in case of actuarially fair premiums and in the absence of altruistic preferences.¹ Privately insured often have a choice between

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¹ See for theoretical models, for example, Refs. [1–6]. For empirical models, see Refs. [7,8].

a low- and a high-option health plan, i.e. a health insurance policy with or without a deductible and with otherwise identical benefits. Of course, this choice will depend on, among others, the level of the premium reduction in case of a deductible. In 1994, about three-quarters of the private health insurers in the Netherlands offered at least one insurance policy with a premium reduction higher than the increase of the deductible amount. One insurer even offered a premium reduction of 1972 guilders in case of a deductible of 1143 guilders ([9], p. 75). Then, the choice between a deductible and full insurance is not difficult. This phenomenon is not restricted to the Netherlands only. Newhouse ([10], p. 1255, note 57) gives an example of high- and low-option plans offered to Harvard employees. These plans differed only in the amount of deductible, but the premium difference exceeded the deductible difference. An intriguing question then is: are such large premium reductions actuarially justified?

In this article we will deal with this question by analyzing the components of the reduction, in case of cost sharing, of the so-called ‘actuarially fair premium’, the latter concept being defined as the insurance premium charged to an individual which equals the expected loss of that individual covered by the insurance policy. Various factors influence this premium reduction: (1) out-of-pocket payments, (2) moral hazard, (3) administrative costs, and, in case of voluntary cost sharing, (4) adverse selection [11]. The purpose of this research is to estimate these partial (i.e. *ceteris paribus*) effects and the total effect of these factors on the actuarially fair premium reduction. We consider a basic health insurance that covers approximately the cure sector completely, apart from dental care. The long-term care sector is not considered here. Another limitation is that in this study we concentrate on deductibles only. For other types of cost sharing the premium reductions can be estimated analogously.

Section 2 investigates the partial effects of the aforementioned factors, which influence the actuarially fair premium reduction in case of a deductible. In Section 3 the total effect of these factors is estimated by means of a premium model. Finally, Section 4 summarizes and discusses the results of this model.

2. Partial effects

Firstly, the partial effects of three factors on the actuarially fair premium reduction in case of health insurance with a *compulsory* deductible are investigated, namely (1) out-of-pocket payments, (2) moral hazard, and (3) administrative costs. Subsequently, the effect of (4) adverse selection as a consequence of health insurance with a *voluntary* deductible is estimated.

2.1. Out-of-pocket payments

In case of a deductible the health care costs in a certain year up to the deductible amount are on account of the insured, and health care costs above that amount are paid by the insurer. This is one of the reasons why the actuarially fair premium for an insurance with a deductible is lower than the actuarially fair premium for full insurance.

We have determined the out-of-pocket expenditure (per person and per year) for various amounts of deductible, ignoring the effect of moral hazard and adverse selection. This out-of-pocket expenditure has been calculated with 1993-data of 10 553 Dutch sickness fund insured.² In the period concerned, these insured had the same comprehensive health insurance policy nearly without any copayments. With these data we have simulated the individual out-of-pocket expenditure as if a deductible would have been introduced. So, the partial effect of out-of-pocket payments on premiums is determined. The simulated mean out-of-pocket expenditure can be easily calculated if the empirical distribution of individual health care costs is known (apart from moral hazard and adverse selection). The out-of-pocket expenditure as a result of a deductible is obtained by truncating the empirical health care costs distribution at the amount of the simulated deductible. We also estimated the simulated out-of-pocket expenditure by means of a two-part model [13,14]. Then, health care costs are modelled into two parts, namely in: (1) the probability of having any costs in a year; and (2) the non-zero costs, which are assumed to be lognormally distributed [15]. The main advantage of such a two-part model is that with the estimation of only three parameters the out-of-pocket expenditures can be

Table 1

Mean (empirical) and estimated expected (two-part model) out-of-pocket expenditure and standard deviation per individual deductible for a basic health insurance policy, apart from moral hazard and adverse selection (guilders, 1993)

Deductible (<i>d</i>)	Mean out-of-pocket expenditure (standard deviation)		Expected out-of-pocket expenditure, two-part model (standard deviation)		Difference (expected – mean)	
0	0	(0)	0	(0)	0	(0)
50	42	(17)	42	(18)	0	(1)
100	78	(37)	79	(37)	1	(0)
200	142	(79)	143	(79)	1	(0)
300	196	(122)	198	(122)	2	(0)
400	244	(166)	246	(165)	2	(–1)
500	288	(208)	289	(207)	1	(–1)
600	328	(251)	329	(249)	1	(–2)
700	365	(291)	365	(290)	0	(–1)
800	399	(331)	398	(329)	–1	(–2)
900	430	(371)	429	(368)	–1	(–3)
1000	460	(409)	458	(406)	–2	(–3)
1500	584	(587)	580	(583)	–4	(–4)
2000	679	(746)	675	(743)	–4	(–3)
3000	823	(1025)	817	(1022)	–6	(–3)
4000	930	(1269)	922	(1262)	–8	(–7)
∞	1800	(6488)	1800	(8843)	0	(2355)

² See Lamers [12] for a detailed description of these data.

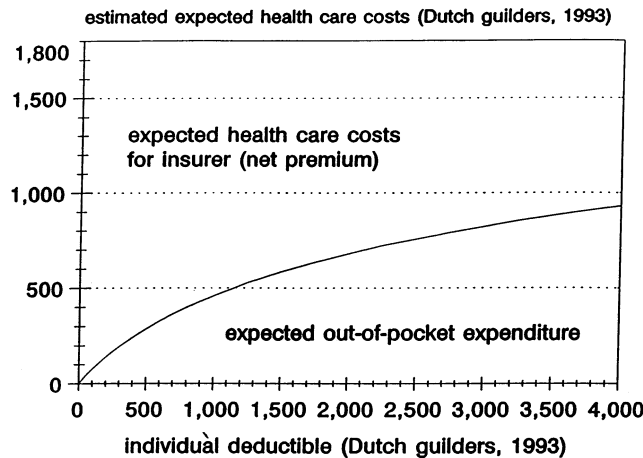


Fig. 1. Estimated expected out-of-pocket expenditure and health care costs for insurer in relation to an individual deductible, apart from moral hazard and adverse selection (source: Table 1).

estimated for any deductible amount as well as for coinsurance with a maximum (and combinations of both).³

In Table 1 both the mean empirical and the expected out-of-pocket expenditure are shown, apart from moral hazard and adverse selection.⁴ For common amounts of the deductible the results are virtually the same.⁵ Therefore, we continue our analysis with the expected out-of-pocket expenditure.

As a result of the introduction of a compulsory individual deductible of 200 guilders per year in a basic health insurance policy the estimated expected out-of-pocket expenditure equals 143 guilders⁶, apart from moral hazard.⁷ The marginal

³ The formulas are available from the authors.

⁴ The mean health care costs of the empirical distribution is 1800 guilders. The two-part model gives — with $\hat{\rho} = 0.877$, $\hat{\mu} = 6.087$, and $\hat{\sigma} = 1.757$ — expected health care costs of 1809 guilders. The expected health care costs and out-of-pocket expenditure are corrected for this difference by setting the expected health care costs equal to 1800 guilders (see Ref. [16], for the method used).

⁵ For other coverages, for example a deductible specific for drugs, dental care, or technical aids, the expected out-of-pocket expenditures also nearly equal the mean empirical out-of-pocket expenditure, apart from moral hazard and adverse selection [17].

⁶ One Dutch guilder was worth about 0.5 US\$ (July 1998).

⁷ The expected out-of-pocket expenditure as a result of coinsurance with an upper limit on out-of-pocket expenditure can be derived from the expected out-of-pocket expenditure in case of a deductible. For example, a 25%-coinsurance rate for the first 800 guilders of costs, implying an upper limit on out-of-pocket expenditure of 200 guilders, results in an expected out-of-pocket expenditure equal to 25% of the out-of-pocket expenditure in case of a deductible of 800 guilders. In this study such a 25%-coinsurance plan results in an expected out-of-pocket expenditure of 100 guilders (Table 1, 100 guilders $\approx 0.25 \times 398$ guilders). Generally, the expected out-of-pocket expenditure (and actuarially fair premium reduction) increases if the coinsurance rate increases, assuming the same upper limit on out-of-pocket expenditure, and apart from moral hazard and adverse selection. Hence, the maximum financial loss for an insured as a result of cost sharing, i.e. the maximum amount of out-of-pocket expenditure minus the premium reduction, is larger the lower the coinsurance rate given the same maximum amount of out-of-pocket expenditure.

increase of expected out-of-pocket expenditure declines as the deductible increases (Fig. 1). On the individual level, the standard deviations of the out-of-pocket expenditure are large. However, a sample of 10 553 insured — as in our data set — results in a 95% confidence interval of the out-of-pocket expenditure equal to $<450 \leftrightarrow 466>$ guilders in case of an individual deductible of 1143 guilders. Thus, the effect of out-of-pocket expenditure on the average actuarially fair premium reduction is surrounded with little uncertainty for an insurer.

The gradient of Fig. 1 reflects the expected percentage of insured who have costs above the deductible.⁸ This fraction is depicted in Fig. 2. In case of a hypothetical deductible of 0 this percentage is equal to the probability of having any health care costs in a year. In our data set this probability is 0.877. In case of a deductible of 200 guilders 58.9% of the insured pay this amount out of their own pocket. The curve is completely flat at a deductible of infinite (no insurance): nobody makes more health care costs than such an amount. Furthermore, from Fig. 2 the distribution of the individual out-of-pocket expenditure can be deduced for each deductible amount. For example in case of a deductible of 200 guilders, 12.3% of the insured has no out-of-pocket expenditure while 58.9% pass the maximum of 200 guilders. So, in-between values of out-of-pocket expenditure have a frequency of 28.8%: 9.5% between 0 and 50 guilders; 8.1% between 50 and 100 guilders; and 11.2% between 100 and 200 guilders.

The out-of-pocket expenditure is related to various individual risk factors. For the basic health insurance policy considered here the out-of-pocket expenditure is in

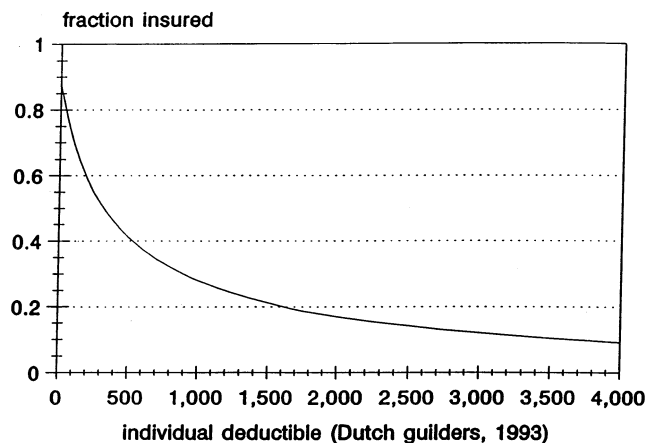


Fig. 2. Fraction insured with health care costs above deductible, apart from moral hazard and adverse selection (average costs equal 1800 guilders).

⁸ A proof is available from the authors.

general lower for the healthy and young people than for the sick and elderly. For men between 5 and 25 years of age the expected out-of-pocket expenditure in case of a deductible of 200 guilders per person per year equals 99 guilders, while for women of 65 years and older it equals 182 guilders. Of course, a more differentiated expected out-of-pocket expenditure (and expected health care costs paid by the insurer) — apart from moral hazard and adverse selection — can be calculated if in addition to age and gender other risk factors are considered as well.

Summarizing, with a two-part model expected out-of-pocket expenditures can be estimated which proved to be nearly equal to the mean empirical out-of-pocket expenditures. The expected out-of-pocket expenditure is surrounded — apart from moral hazard and adverse selection — with little uncertainty. So, at portfolio level insurers can fairly accurately estimate the effect of out-of-pocket payments on the actuarially fair premium reduction.

2.2. *Reduction of moral hazard*

Health insurance may result in moral hazard, i.e. insured have less incentives to avoid a health care costs episode and, in case of such an episode, to restrain these costs as much as possible [18]. The emphasis in the definition of moral hazard changed slightly over the years, from immoral behavior (e.g. fraud) to rational behavior (pure price-effect).⁹ For health insurance the latter aspect of moral hazard seems to dominate.

Inasmuch as cost-sharing reduces moral hazard, the actuarially fair premium reduction in case of cost-sharing should reflect this reduced moral hazard. In order to estimate this component of the premium reduction in case of a deductible, we need reliable estimates of the effect of a deductible on an individual's total annual health expenses. Although many studies conclude that an increase of the level of the out-of-pocket payments causes a decline of the demand for the health care concerned [21–23], the exact degree of this response is uncertain. Because of methodological and practical problems [24–26] the results of most of these studies are surrounded with great uncertainty. Because the RAND Health Insurance Experiment was specifically designed to overcome these methodological and practical problems [27], we use the results of that study to calculate the moral hazard component for our premium model.

According to the RAND-experiment, people who were fully insured had 46% more costs than those with a high deductible (up to \$1143). Furthermore, an individual deductible of \$150 specific for non-clinical health care (and with hospital care free) resulted in a reduction of 8% of the hospitalization probability and in a reduction of 20% of the total health care costs compared to full coverage. Besides, the RAND-experiment shows that cost sharing mainly influences the number of health care costs episodes, but hardly affects the costs of such an episode [27]. Keeler et al. [28] developed a simulation model with parameter estimates based on

⁹ See for a discussion Refs. [1,19,20].

the RAND-experiment. That model gives quite remarkable results. For example, an individual deductible of 200 dollar results in a reduction of health care costs of about 200 dollar per person. Then, an insurer can offer a deductible of 200 dollar with a premium reduction — based on only moral hazard — of about 200 dollar. Consequently the insured bear no financial risk when they choose such a policy compared to full health insurance.

The partial effect of out-of-pocket payments on the actuarially fair premium reduction can probably be estimated more accurately than the partial effect of moral hazard. Moreover, the moral hazard effect cannot easily be determined for different insurance coverages and types of cost sharing, in contrast to the expected out-of-pocket expenditure.

2.3. Administrative costs

Insurers have to add a loading fee to the net premium because of administrative costs. Thorpe [29] distinguished the following components of the loading fee: (1) transaction-related costs; (2) overhead costs; (3) selling and marketing costs; (4) regulatory and compliance costs; and (5) profit. Theoretically, these components primarily result in a fixed loading fee and/or a loading fee proportional to the expected health care costs.

In the Dutch sickness fund insurance, health care providers submit their bills directly to the insurer of their patients. On the other hand, privately insured pay the bill in general themselves and get reimbursed afterwards. In practice, administrative costs are for public health insurance lower than for private health insurance [30].

Of course, a fixed loading fee component is independent of whether or not an insurance plan has a deductible. In private health insurance, the transaction-related administrative costs decline in case of a deductible. On the other hand, in a reimbursement system like the Dutch sickness fund insurance these costs will increase in case of the introduction of a deductible, apart from moral hazard, owing to the fact that either insurers or health care providers have to collect the out-of-pocket payments.

In conclusion, the administrative costs for an insurer strongly depend on the health care system (public or private insurance). This makes the effect of a deductible on administrative costs uncertain. Therefore, we will not take these costs explicitly into account in the premium model.

2.4. Adverse selection

Hitherto we discussed the three factors which affect the actuarially fair premium reduction in case of a *compulsory* deductible. In case of a *voluntary* deductible a fourth relevant factor is adverse selection, i.e. the selection that occurs as a result of the tendency of high-risk individuals to be more likely to buy a certain insurance coverage than low-risk individuals within the same premium risk group. For the effect of adverse selection on health insurance premiums we used the same costs data as were used to estimate the out-of-pocket expenditures. Furthermore, we used

survey data of the same 10 553 sickness fund insured. These insured, who were hardly confronted with copayments at that time, have been asked whether they wished to conclude supplementary insurance for an appropriate premium if a compulsory deductible specific for drugs would have been introduced.¹⁰ A total of 51.5% of the respondents answered that they would like to have supplementary insurance covering out-of-pocket payments. Univariate analyses (not shown) indicate that mainly high-risk individuals are inclined to conclude this supplementary insurance. According to multivariate analyses, the choice is especially related to last year's medical consumption and to perceived health status. The health perceptions 5 years ago, and the expectations of the insured with regard to hospitalizations and visiting a medical specialist next year did not have statistically significant effects on the choice for supplementary insurance (*ceteris paribus*).

The annual health care costs data concerned the period of about 2 months before and 10 months after the survey was conducted. The mean annual health care costs were equal to 1366 guilders for insured who were not inclined to conclude supplementary insurance (low-option group) versus 2210 for those who favored the supplementary insurance (high-option group) (+ 62%).¹¹ This means a substantial potential adverse selection. For the analysis of the adverse selection effect in case of a voluntary deductible, we assume that those in the low-option group prefer the deductible and those in the high-option group prefer full coverage.

An important question is whether an insurer can prevent adverse selection by risk-rating the premiums. Van de Ven and Van Vliet [11] developed an indicator of the extent of consumer information surplus (*vis-à-vis* an insurer) with regard to future health care costs. They used the following definition: 'the difference in average *actual* total future expenditures — apart from moral hazard — between the high-option group and the low-option group, *minus* the difference in average *predicted* total future expenditures between the two groups, with the predictions based on the relevant risk factors.' In formula:

$$\text{CIS}(X) = [\bar{Y}_H - \bar{Y}_L] - [\bar{\hat{Y}}_H(X) - \bar{\hat{Y}}_L(X)]$$

where $\bar{Y}_H$ and $\bar{Y}_L$ are the average *actual* total future expenditures in the high- and low-risk group, respectively, and $\bar{\hat{Y}}_H(X)$ and $\bar{\hat{Y}}_L(X)$ are the average *predicted* total future expenditures in these groups based on the risk factors X . The better X captures the real risk factors, the closer the predicted values are to the actual values, for both the high- and low-risk groups, and thus the smaller is the consumer information surplus, $\text{CIS}(X)$. Given a premium structure based on X , a CIS equal

¹⁰ At the time of the survey this was an important policy issue. Because the survey question concerned a deductible specific for drugs we have to assume that this choice will be a good proxy of such a choice if a deductible would be introduced for the whole basic benefits package.

¹¹ Van de Ven and Van Vliet ([11]) used a comparable survey question to investigate the effect of adverse selection for privately insured. In that study, 1 year after the survey choice, the high-option group had 57% higher health care costs than the low-option group.

Table 2

Consumer information surplus, basic health insurance (guilders, 1993)

Model	Mean predicted costs (1800 guilders)		Consumer information surplus	
	Low-option group (mean 1366)	High-option group (mean 2210)	Guilders (std. error)	Index
1 no risk factors	1800	1800	844 (127)	100**
2 age/gender	1685	1909	619 (125)	73**
3 age/gender, demographic risk factors ^a	1654	1939	558 (124)	66**
4 age/gender, expected health care use ^b	1558	2030	371 (121)	44**
5 age/gender, perceived health status	1535	2050	328 (122)	39**
6 age/gender, ln(costs + 1) '88–'92	1460	2120	184 (114)	22

^a The demographic risk factors are: education level, source of income, degree of urbanization, marital status, and whether or not having another supplementary health insurance.

^b The expectation of the insured with regard to a hospital admission and with regard to a consultation with a medical specialist next year. Of course, an insurer does not know these expectations.

** $P < 0.01$.

to zero means that with such risk factors adverse selection is avoidable for an insurer.

In Table 2 the consumer information surplus is shown as a function of several sets of risk factors. Without any risk factor — a flat premium rate (community rating) — the predicted expenditures for the low- and high-option group equal 1800 guilders. Because the mean expenditures in the low- and high-option groups are 1366 and 2210 guilders, respectively, the consumer information surplus in case of risk-independent premiums equals 844 guilders. An age-gender-rated premium (see model 2, Table 2) decreases the consumer information surplus with 225 guilders (– 27%). If the premiums are related to age, gender and individual health care costs in the past (model 6), CIS is reduced by 78% (to 184 guilders) and is not statistically significant greater than 0. This indicates that insurers can largely avoid adverse selection by adequate risk-rating.

A limitation of our study is that adverse selection is only related to a situation in which consumers had a survey choice whether or not to conclude supplementary insurance. However, it is not an unrealistic assumption that consumers will make the same choice in practice (see for example Refs. [31,32]). On the other hand, an advantage of a survey choice is that the partial adverse selection effect can be investigated. In case of a real choice one observes both the effect of adverse selection and moral hazard. Another limitation of our study is that the questionnaire mentioned neither the deductible amount nor the premium for supplementary

insurance. Furthermore, the effect of adverse selection has been investigated with health care costs data in the period of about 2 months before and 10 months after the survey choice. So the consumer information surplus has probably been overestimated in this study.¹²

3. Premium model

3.1. Compulsory cost sharing

In the absence of moral hazard, the actuarially fair premium reduction in case of compulsory cost sharing is equal to the expected out-of-pocket expenditures, apart from administrative costs. Then, the estimated premium reductions in Section 2.1

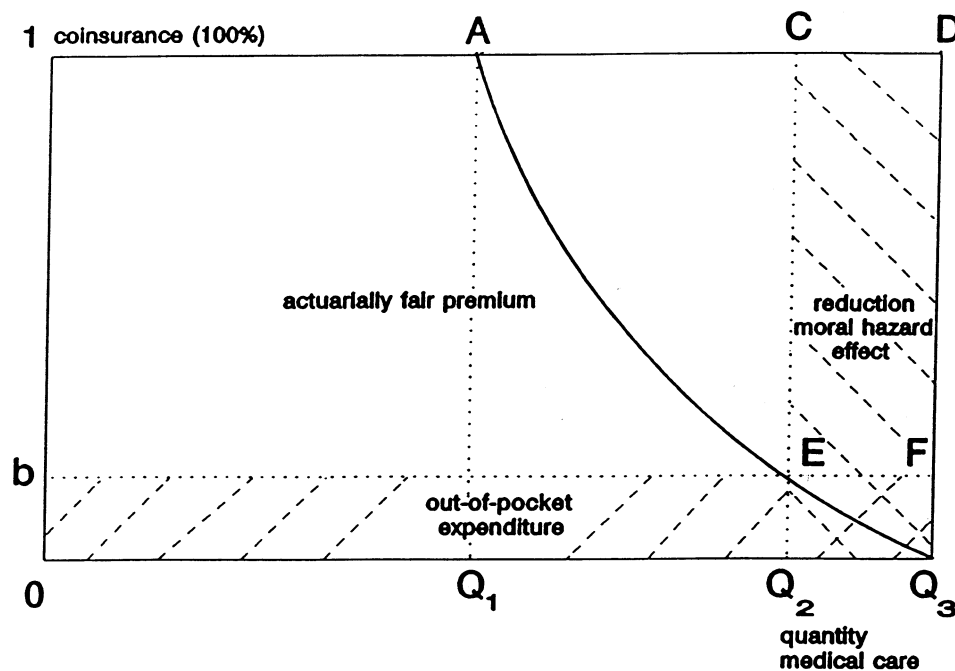


Fig. 3. Out-of-pocket expenditure, moral hazard, and actuarially fair premium for a coinsurance plan without an upper limit, with AEQ_3 an individual demand curve.

¹² With data of only 10 months after the survey choice we cannot investigate the effect of regression to the mean. However, according to a study of Van de Ven and Van Vliet ([11]) the cost difference between the high- and low-option group was roughly stable during a period of 8 years after the survey choice. This suggests that with such a survey choice the effect of regression-to-the-mean is minimal, i.e. that the cost difference between the high- and low-option group persists over time.

are sufficient for a premium model. However, the existence of moral hazard complicates the calculation of actuarially fair premium reductions.

The relationship between the expected out-of-pocket expenditure and the moral hazard effect is of importance to determine actuarially fair premium reductions for policies with compulsory cost sharing (apart from administrative costs). In case of coinsurance ‘without’ an upper limit this relationship is depicted in Fig. 3 [28]. Without loss of generality, the price of one unit of medical care is set equal to 1. Curve AEQ_3 is a demand curve of an ‘average’ individual. The expected demand is equal to Q_2 at a coinsurance rate of b . Then, the expected health care costs are equal to $01CQ_2$ and the reduction of the moral hazard effect compared to full insurance is equal to Q_2CDQ_3 . The expected out-of-pocket expenditure corresponds to $0bEQ_2$ and the actuarially fair premium is equal to $(1 - b) \cdot Q_2$ (rectangle $b1CE$).

The premium reduction in case of a coinsurance rate of b is made up of two components: on the one hand the expected out-of-pocket expenditure (bottom bar), and on the other hand the reduction of the moral hazard effect (right column). Rectangle Q_2EFQ_3 is equal to the expected reduction of the out-of-pocket expenditure because of a reduction of moral hazard (the interaction effect between cost sharing and moral hazard). So, in case of coinsurance *without* an upper limit the moral hazard effect can simply be divided into an effect on the out-of-pocket expenditure and an effect on the health care costs for account of the insurer. Theoretically, such a relationship does not exist for deductibles and for coinsurance plans *with* an upper limit on out-of-pocket expenditure. Therefore, for such insurance plans we have to estimate this relationship empirically.

The moral hazard effects used in our premium model are based on the RAND-experiment (Section 2.2). On the basis of these effects ([28], pp. 104–105) we estimated a log-linear function between costs per plan and the deductible amount. This method provided us with moral hazard effects for every deductible level (comparable with Q_2CDQ_3 , Fig. 3). Keeler et al. [28] have also simulated insurance plans with a maximum out-of-pocket expenditure of 50 and 100 dollar. We did not estimate the moral hazard effects for deductibles below 150 dollar because the maximum level on out-of-pocket expenditure during the RAND-experiment was at least 150 dollar (individual deductible plan). Although the participants of the RAND-experiment hardly anticipated on reaching the maximum amount of out-of-pocket expenditure ([27], pp. 114–117), it remains questionable whether the same is true for a maximum as low as 50 or 100 dollar.

Furthermore, we have to make an assumption about the interaction effect between moral hazard and out-of-pocket expenditure. If we assume that an expected cost-reduction takes place exclusively via a decrease of the probability of having any health care costs — which seems a reasonable assumption ([27], p. 44) — the cost-reduction is the same percentage for the insured (out-of-pocket expenditure) and for the insurer (claims).¹³ A low deductible results in a relatively small expected out-of-pocket expenditure compared to the total health care costs. This

¹³ With this assumption, the interaction effect between moral hazard and out-of-pocket expenditure in case of a deductible corresponds with that for coinsurance *without* an upper limit (Fig. 3).

Table 3

Moral hazard effect (exclusively via probability of having costs), expected out-of-pocket expenditure and expected health care costs above deductible (guilders, 1993)

Individual deductible	Cost reduction because of less moral hazard	Expected out-of-pocket expenditure (adjusted for moral hazard)	Expected health care costs above deductible (net premium)
0	0	0	1800
300	463	147	1190
400	489	179	1132
500	508	208	1084
600	525	233	1043
700	538	256	1006
800	549	277	974
900	559	296	945
1143	568	313	919
1500	602	386	812
2000	625	440	735
2500	643	483	674
3000	657	519	624
3500	669	548	583
4000	680	574	546
∞	767	1033	0

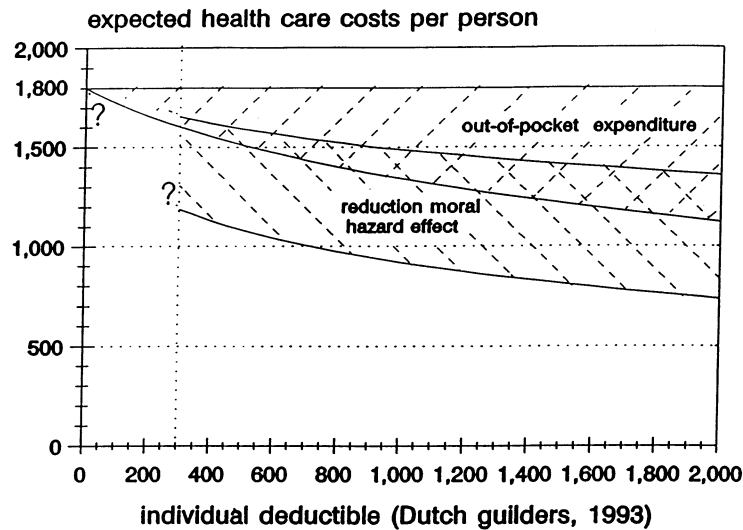


Fig. 4. Expected out-of-pocket expenditure and reduction of moral hazard per individual deductible (source: Table 3).

implies that the reduction of moral hazard in case of a low deductible is reflected for a small part in the expected out-of-pocket expenditure and for a substantial part in the expected health care costs above the deductible (net premium).

The expected out-of-pocket expenditure adjusted for a reduction of the moral hazard effect is depicted in Table 3 (see also Fig. 4). In case of low deductibles, these out-of-pocket expenditures differ little (in absolute terms) from the out-of-pocket expenditures of Table 1 (no moral hazard adjustments). For example, these amounts are, respectively, 147 and 198 guilders for a compulsory deductible of 300 guilders. Then, the total reduction of moral hazard is estimated to be 463 guilders. So, with a deductible of 300 guilders an insurer can expect — compared to full insurance — 147 guilders lower health care costs because of out-of-pocket payments ($ObEQ_2$, Fig. 3), plus a decrease of 412 guilders because of a reduction of moral hazard ($ECDF$), plus a reduction of 51 ($= 198 - 147$) guilders due to the interaction between moral hazard and cost sharing (Q_2EFQ_3).¹⁴

The premium model that takes into account both the expected average out-of-pocket expenditure and moral hazard effect gives for deductibles between 300 and 800 guilders an average premium reduction which is larger than the deductible itself. Such a combination of deductible and premium reduction would imply that insured can never have a financial loss compared to full coverage.

3.2. Voluntary cost sharing

In case of voluntary cost sharing adverse selection may occur. Firstly, the effect of adverse selection on the expected out-of-pocket expenditure and on the health care costs for the insurer (net premium) is estimated, apart from the moral hazard effect. Secondly, the interaction effect between adverse selection and moral hazard is considered.

The effect of adverse selection depends on — among others — the extent to which premiums are risk rated. With sufficiently risk-rated premiums an insurer can prevent adverse selection (Table 2). Then, the actuarially fair premium structure will be the same as in case of compulsory cost sharing (Section 3.1). For simplicity, in this section we assume risk-independent premiums. The difference in health care costs between the high- and low-option group is then, apart from the moral hazard effect, equal to 844 ($= 2210 - 1366$) guilders. So, theoretically, insured who belong to the low-option group can receive an actuarially fair premium reduction of 844 guilders in case of a voluntary deductible of 0 guilders, relative to the premium for full insurance.¹⁵ For ‘other’ levels of deductible the premium reduction consists of two components, namely adverse selection and the expected out-of-pocket expenditure (apart from moral hazard and administrative costs). As can be expected, the out-of-pocket expenditure for the low-option group will be lower than the (hypothetically) out-of-pocket expenditure for the high-option group, assuming the same

¹⁴ If the cost-reduction is achieved exclusively via the positive health care costs, the interaction effect between cost sharing and moral hazard is smaller. Then, the estimated premium reductions will be larger.

¹⁵ Actually, fully insured are confronted with a premium rise of 410 ($= 2210 - 1800$) guilders and those with a deductible receive a premium reduction of 434 ($= 1800 - 1366$) guilders, as compared to community rated premium in case of no consumer choice for a deductible-option.

deductible amount. Table 4 shows the expected average out-of-pocket expenditure and the expected average health care costs for account of the insurer (net premium). Generally, adverse selection has a small absolute effect on the expected out-of-pocket expenditure. Like moral hazard, adverse selection has mainly consequences for the expected health care costs above the deductible amount (net premium).

Hitherto, in case of a voluntary deductible moral hazard is not considered. In the RAND-experiment, the proportional cost-reduction due to less moral hazard as a result of a deductible was the same for the healthy and the sick ([27], pp. 48–49). Generally, low-risk individuals (the healthy) prefer a deductible. Hence, in absolute terms, the expected reduction of moral hazard per person with a deductible in case of a voluntary deductible will be less than in case of a compulsory deductible. We assume that the moral hazard effect in case of a voluntary deductible is equal to 1366/1800 of the moral hazard effect at the same amount of compulsory deductible (with 1366 the mean health care costs for the low-option group, and 1800 for the complete sample).

The mean premium reduction in case of a voluntary individual deductible of 1500 guilders per year could be about 1500 guilders per person (Fig. 5). This reduction is based on the expected out-of-pocket expenditure, moral hazard, and adverse selection. Consequently, one could then choose a deductible of 1500 guilders

Table 4

Expected health care costs up to and above the deductible amount for the low- and high-option group, apart from moral hazard (guilders, 1993)

Individual deductible	Expected costs low-option group (48.5%)		Expected costs high-option group (51.5%)	
	Up to deductible (out-of-pocket)	Above deductible (net premium)	(Hypothetically) Up to deductible	Above deductible (net premium)
0	0	1366	0	2210
50	40	1326	43	2167
100	74	1292	83	2127
200	133	1233	152	2058
300	182	1184	212	1998
400	225	1141	266	1944
500	263	1103	314	1896
600	296	1070	359	1851
700	328	1038	400	1810
800	356	1010	438	1772
900	382	984	474	1736
1143	406	960	507	1703
1500	506	860	649	1561
2000	582	784	761	1449
2500	643	723	854	1356
3000	695	671	932	1278
3500	738	628	1000	1210
4000	775	591	1060	1150
∞	1366	0	2210	0

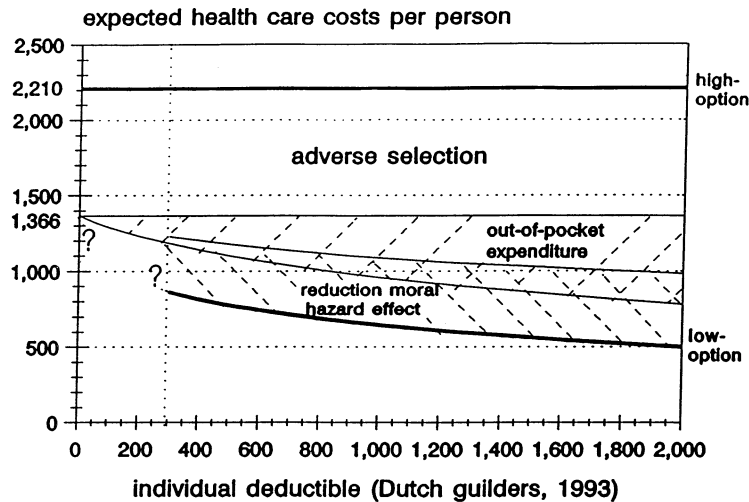


Fig. 5. The effect of out-of-pocket payments, moral hazard, and adverse selection on the average actuarially fair premium (individual deductible; guilders, 1993).

without any financial risk compared to full coverage. Of course, the effect of adverse selection on the actuarially fair premium reduction decreases to the extent that the premium for the basic insurance is risk rated.

4. Conclusion and discussion

4.1. Conclusion

Theoretical and empirical studies suggest that in case of health insurance with an actuarially fair premium structure some cost sharing is 'optimal' for an insured. This article investigates which actuarially fair premium reduction insured can receive in case of a deductible compared with full insurance with otherwise identical benefits. Various factors influence this premium reduction: (1) out-of-pocket expenditure, (2) moral hazard, (3) administrative costs, and, in case of a voluntary deductible, (4) adverse selection. We estimated these partial effects and their total effect on the actuarially fair premium reduction.

With a two-part model the expected annual out-of-pocket expenditure is estimated for insured who had full insurance coverage. So, we answered the question what these insured would have paid out-of-pocket if a compulsory deductible had been introduced, apart from moral hazard. These estimates proved to be virtually equal to the mean simulated empirical out-of-pocket expenditure. For a basic health insurance the expected out-of-pocket expenditure are 143 guilders in case of a deductible of 200 guilders (apart from moral hazard and adverse selection). The quotient of expected out-of-pocket expenditure and deductible declines as the

amount of deductible increases. For example, in this case the expected out-of-pocket expenditure equals 458 guilders for a deductible of 1143 guilders. Then, insured can receive an actuarially fair premium reduction of on average 458 guilders, apart from other effects.

The simulated effect of a reduction of moral hazard as a result of a deductible is based on the results of the RAND-experiment. The reduction of moral hazard appears to have a small effect on the expected out-of-pocket expenditure but a substantial effect on the expected health care costs above the deductible (premium).

The effect of a deductible on administrative costs is difficult to estimate. In a reimbursement system these costs will decline as a consequence of the introduction of a deductible. On the other hand, a deductible may increase administrative costs if health care providers submit their bills directly to the insurer of their patients.

According to our premium model the actuarially fair premium reduction in case of a *compulsory* deductible between 300 and 800 guilders is higher than the deductible amount itself (premium reduction based on out-of-pocket expenditure and moral hazard reduction).¹⁶ Then, insured could never have a financial loss compared to full insurance.¹⁷

In case of a *voluntary* deductible adverse selection may occur, and consequently the premium reduction in case of a deductible may further increase. We have analyzed this adverse selection effect on the basis of a survey question about the inclination to take supplementary insurance in case of the introduction of a compulsory deductible. The mean health care costs were 1366 guilders for insured who were not inclined to conclude the supplementary insurance (low-option group) versus 2210 for those who favored the supplementary insurance (the high-option group) (+ 62%). This means a substantial potential adverse selection. So, theoretically, insured who opt for a deductible can receive — only because of adverse selection — a premium reduction of 844 guilders compared to the premium for full insurance if this premium is not risk rated. Like moral hazard, adverse selection has a small effect on the expected out-of-pocket expenditure but a substantial effect on the expected health care costs paid by the insurer (net premium). The extra premium reduction because of adverse selection depends on the risk factors insurers use for determining the premium for full insurance. According to our study insurers can for the greater part reduce adverse selection if they ask premiums based on age, gender and individual health care costs in the past.¹⁸

4.2. Discussion

4.2.1. Premium reduction in practice

The average actuarially fair premium reductions estimated with the premium model have been compared with the premium reductions which Dutch private

¹⁶ The overall mean per capita annual expenditures are 1800 guilders.

¹⁷ Because we have taken actuarially fair premiums as a starting point, the effect of employers' contributions to health insurance premiums is not considered.

¹⁸ For similar results, see Ref. [11].

health insurers offer in case of a voluntary deductible. In 1994, about three-quarters of these insurers offered at least one insurance policy with a premium reduction higher than the corresponding increase of the deductible amount. This phenomenon mainly occurred with regard to the premium structure for elderly singles and ‘elderly’ families. These risk-groups have an expected out-of-pocket expenditure close to the deductible amount in case of a low deductible (Section 2.1). Then, a small reduction of moral hazard will be enough to have an actuarially fair premium reduction that exceeds the deductible amount. According to our premium model the reduction of moral hazard is the largest for the high-risk groups. This effect explains why mainly for the elderly premium reductions exceeding the deductible occur. Furthermore, in practice premiums are in general only age-related. So, adverse selection may also be a factor of importance (Section 2.4).

4.2.2. Reliability

To determine the average actuarially fair premium reduction in case of a deductible we used estimated effects with different levels of reliability. We saw that the mean *expected* out-of-pocket expenditure is surrounded — apart from moral hazard and adverse selection — with relatively little uncertainty. So, insurers can fairly accurately estimate this effect on the mean actuarially fair premium (portfolio level). The simulated effect of moral hazard is based on the results of the RAND-experiment. Therefore, we have to make the assumption that the reduced moral hazard is not compensated by supplier-induced demand.¹⁹ This assumption may not be realistic, especially when a large fraction of the population opts for a deductible. Furthermore, for low deductibles policy makers should take into account that insured may anticipate on reaching the maximum out-of-pocket expenditure. So, the effect of cost sharing on health care costs is beset with some uncertainty.

The effect of adverse selection strongly depends on the degree to which premiums are risk rated. The more the premiums are risk rated, the less adverse selection. In this paper we had to assume that people in reality make the same choice as in the survey we used to investigate the effect of adverse selection. In the survey question, we did not mention a deductible amount and a premium. So, the effect of adverse selection is based on the inclination to take a supplementary insurance that covers out-of-pocket payments as a consequence of a compulsory deductible. Furthermore, if *everyone* would opt for a deductible rather than for full coverage, there would be no adverse selection at all.

Despite the assumptions we had to make in our study, we conclude that premium reductions that exceed the deductible amount, as observed in practice, may indeed reflect actuarially fair pricing of health insurance.

¹⁹ See for theoretical discussions about supplier-induced demand [33]. For further comments see for example Refs.: [34–36].

4.2.3. *Supplementary insurance*

Our analysis also has implications for the premium of a supplementary insurance that covers the out-of-pocket payments due to a deductible in the basic insurance of a social health insurance system, like the sickness fund insurance in Europe or Medicare and Medicaid in the United States of America. An important finding of our study is that the additional moral hazard that results from the supplementary insurance, appears to have a small effect on the out-of-pocket expenditures, but a substantial effect on the expected health care costs above the deductible, i.e. the costs to be covered by the basic insurance. In as much as the aim of the deductible is to reduce moral hazard, policy makers are confronted with the question how to prevent the negative external effect of the supplementary insurance on the basic insurance. This can be done by confronting those who buy such a supplementary insurance with the additional costs they create for the basic insurance. The results of our study indicate that in that case for a broad range of deductibles the premium for the supplementary insurance will exceed the deductible. Then no rational consumer is willing to buy such a supplementary insurance. Therefore, as an alternative for complicated regulation to internalize the external effects of supplementary insurance into its premium, government could forbid insurers to sell supplementary insurance that compensates for the out-of-pocket payments due to a deductible in the basic insurance of a social health insurance system.

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